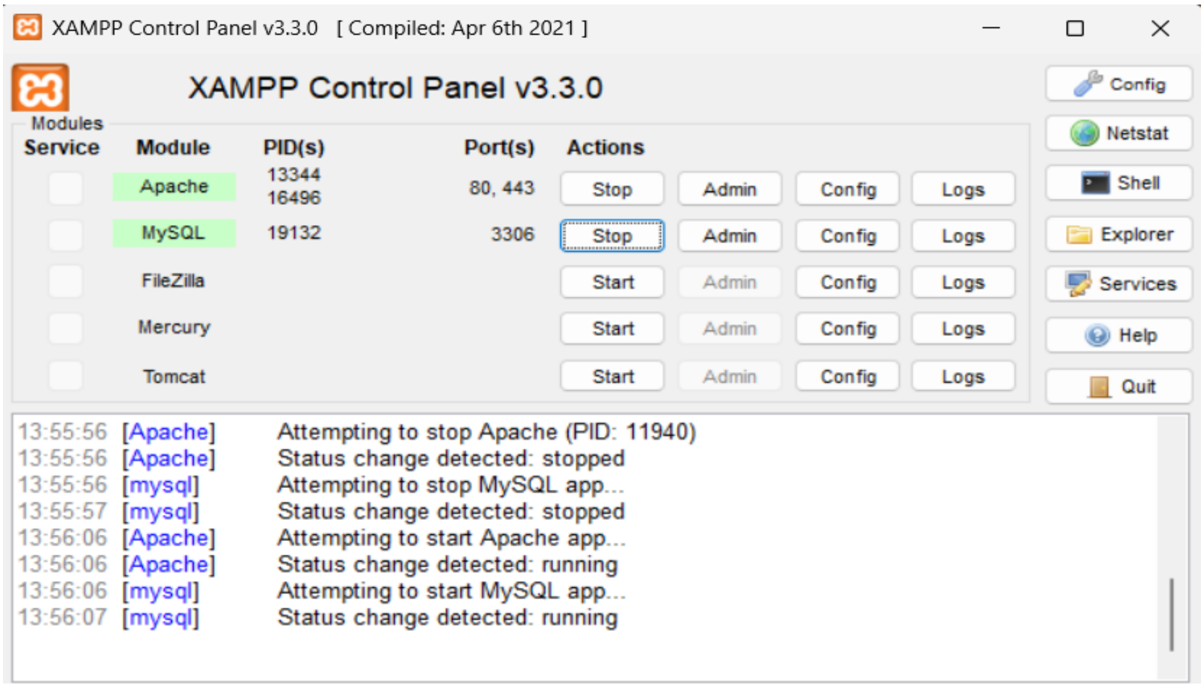
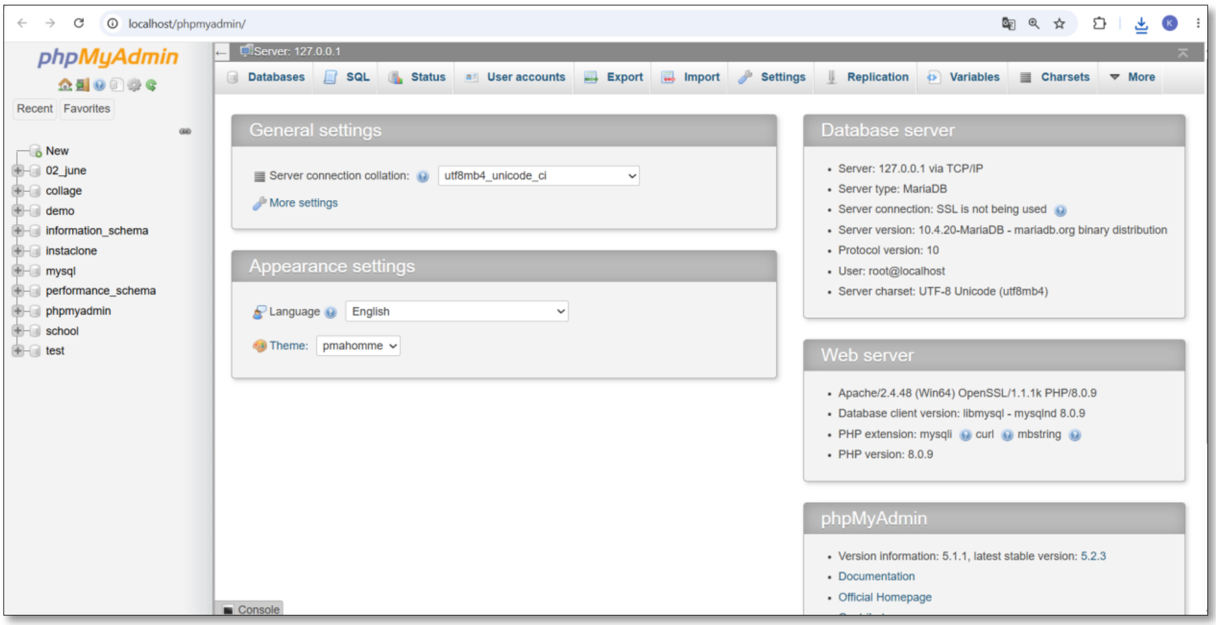
Session 1: Introduction to Databases Trainer Coverage: What is a database, DBMS, RDBMS (with examples). Difference between MySQL, PostgreSQL, Oracle, SQLite. Tables, rows, columns, and relationships. Demo: Show data stored in MySQL Workbench.

<u>TASKS:</u>

Q-1. Install MySQL Workbench on your computer, connect to the local MySQL server, and take a screenshot of the default databases shown in the Schemas panel.

ANS:



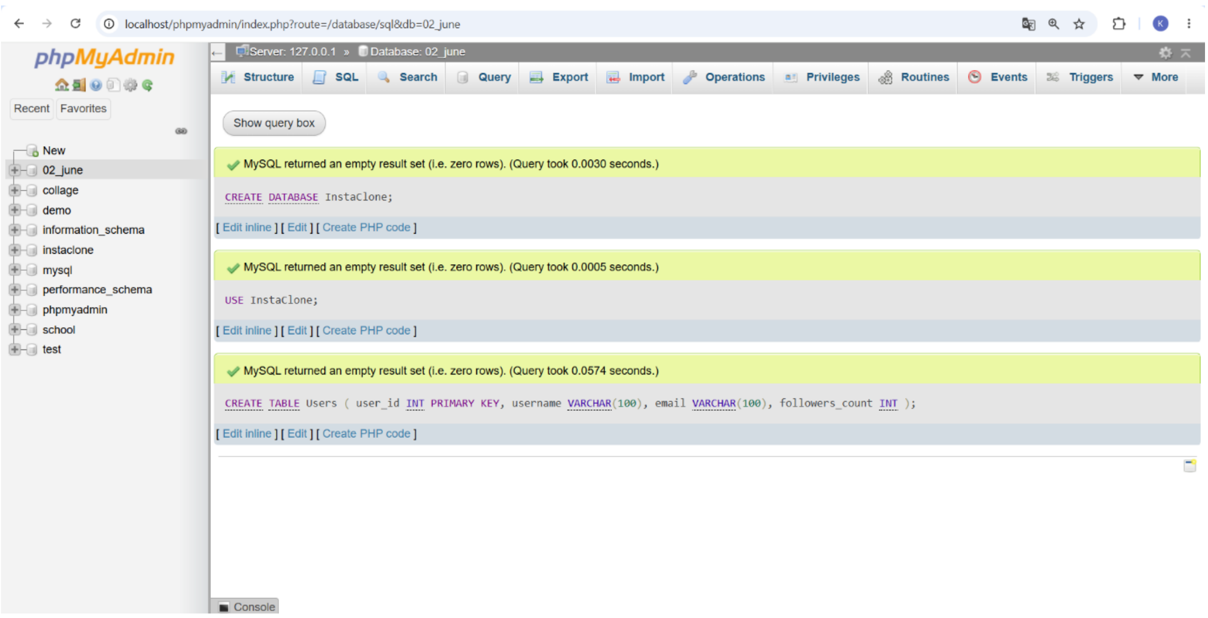


Q-2. Create a new database called InstaClone in MySQL Workbench, then create a table named Users with columns: user_id (INT, primary key), username (VARCHAR), email (VARCHAR), and followers_count (INT).

ANS:

```
USE InstaClone;


CREATE TABLE Users (
   user_id INT PRIMARY KEY,
   username VARCHAR(100),
   email VARCHAR(100),
   followers_count INT
);
```



Q- 3. Insert 3 sample users into the Users table you created, using realistic Instagram-style usernames and follower counts.

ANS:

INSERT INTO Users (user_id, username, email, followers_count)

VALUES

(1, 'krishna_patel', 'krishna@example.com', 1250),

(2, 'travel_with_riya', 'riya@example.com', 8450),

(3, 'tech_by_rahul', 'rahul@example.com', 3200);

Show query box

✓ 3 rows inserted. (Query took 0.0043 seconds.)

```
INSERT INTO Users (user_id, username, email, followers_count) VALUES (1, 'krishna_patel', 'krishna@example.com', 1250), (2, 'travel_with_riya', 'riya@example.com', 8450), (3, 'tech_by_rahul', 'rahul@example.com', 3200);
```

[Edit inline] [Edit] [Create PHP code]

+ Options

←T→				user_id	username	email	followers_count
☐	Edit	Copy	Delete	1	krishna_patel	krishna@example.com	1250
☐	Edit	Copy	Delete	2	travel_with_riya	riya@example.com	8450
☐	Edit	Copy	Delete	3	tech_by_rahul	rahul@example.com	3200

Q-4.Create another table in the InstaClone database called Posts with columns: post_id (INT, primary key), user_id (INT), caption (VARCHAR), and post_date (DATE). Add a foreign key from Posts.user_id to Users.user_id to establish a relationship.

ANS.

USE InstaClone;

CREATE TABLE Posts (

post_id INT PRIMARY KEY,

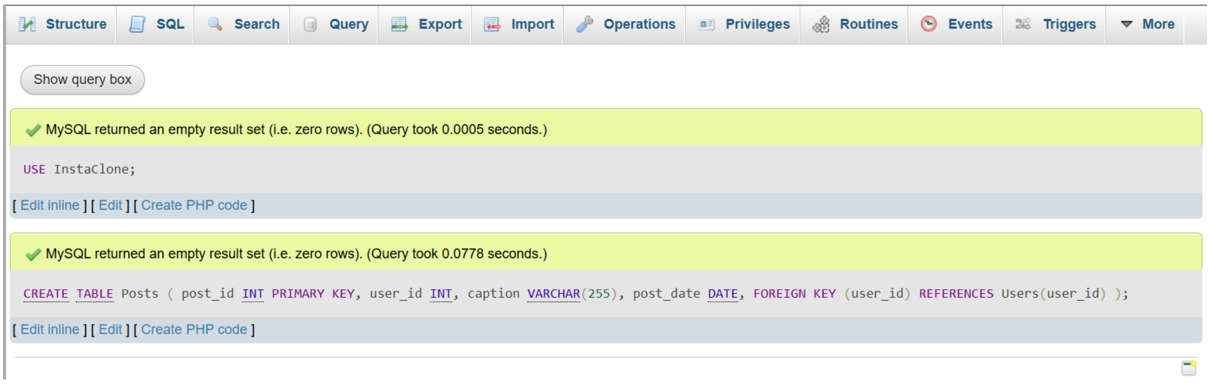
user_id INT,

caption VARCHAR(255),

post_date DATE,

FOREIGN KEY (user_id) REFERENCES Users(user_id)

);



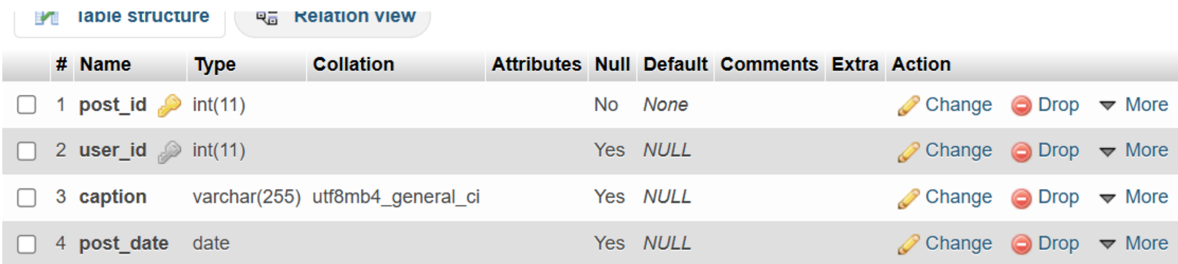


Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	post_id	int(11)			No	*None*			Change Drop More
2	user_id	int(11)			Yes	*NULL*			Change Drop More
3	caption	varchar(255)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
4	post_date	date			Yes	*NULL*			Change Drop More

Q-5. Write a short comparison (3-4 lines each) of MySQL, PostgreSQL, Oracle, and SQLite, focusing on where each one is commonly used (for example: web apps, mobile apps, enterprise systems, etc.).

ANS.

MySQL

MySQL is widely used for web applications and websites.
It is popular with PHP, Java, and other web technologies.
It is commonly used for applications like e-commerce, blogs, and content management systems.

PostgreSQL

PostgreSQL is an advanced open-source relational database.
It is commonly used in web applications, data-intensive applications, and analytics.
It is preferred when complex queries and strong data integrity are required.

Oracle

Oracle Database is mainly used in large enterprise systems.
It is common in banks, insurance companies, government organizations, and large businesses.
It is known for strong security, scalability, and reliability.

SQLite

SQLite is a lightweight, serverless database stored in a single file.
It is commonly used in mobile apps, desktop applications, and embedded systems.
For example, many Android and iOS applications use SQLite for local data storage.